SIMEON JAMES WUTHIER

www.simewu.com ▪ mail@simewu.com

EDUCATION

Ph.D. in Security *Expected Graduation: May 2024*
University of Colorado, Colorado Springs *GPA: 4.0/4.0*

- Recipient of the Scholarship for Service, eligible for SFS Hiring Authority under Public Law 113-274 § 302.
- Developing strategies to counteract networking threats and evaluate communication network protocols.
- Leading research initiatives to enhance the security and reliability of distributed networking architectures.
- Collaborating with interdisciplinary teams on vulnerability identification and security solutions in networked systems.
- Managing security research projects in distributed systems, with results published in esteemed conferences and journals.
- Presenting research findings at domestic and international conferences, and actively participating in IEEE and ACM professional communities.

Master of Science in Computer Science *Expected Graduation: May 2024*
University of Colorado, Colorado Springs

Bachelor of Science in Computer Science *Graduated: May 2020*
University of Colorado, Colorado Springs

- **Honors: President's List** (*Fall 2015, Spring 2016, Spring 2019*)**, Dean's List** (*Spring 2018, Fall 2019*)

TECHNICAL SKILLS & COMPETENCIES

Cryptography: Expertise in traditional applied cryptography and post-quantum techniques, with a focus on protocol design and systems analysis.
- Implemented NIST-standardized post-quantum algorithms in several networking applications, analyzing their impacts, leading to publications.
- Mentored students in six cryptography courses, emphasizing practical usage and implications of the cryptographic principles.

Networking and Communication Protocols: Enhanced QUIC and TCP/IP networks using custom scripts for data analysis. Improved 5G security by implementing additional authentication, tested using srsRAN and software-defined radios; findings were peer-reviewed and published.

Data Mining and Analysis: Applied machine learning to self-extracted technical datasets, utilizing Matlab/Mathematica for data visualization/analysis.

Tools and Platforms: GNU Debugger (gdb), Wireshark/TShark, Scapy, Netcat, Nmap, Metasploit, Z3 Solver, Ghidra, Docker, VMware, Hyper-V, VirtualBox, AWS (EC2), Google Cloud (Compute Engine), GitHub, ZenHub, GitLab.

Languages: C/C++, Python, JavaScript/TypeScript, C#/Visual Basic, Regular Expressions, Unix/Bash, Java, Assembly, Go, PHP, Ruby.

Relevant Coursework: Applied Cryptography, Software Engineering, Data Structures and Algorithms, Operating Systems, Wireless Embedded System Security, Fundamentals of Computer/Network Security, Computational Automata and Formal Languages, Calculus, Discrete Mathematics.

EXPERIENCE

Teaching Assistant – *CS 4920 / 5920*, Applied Cryptography *Jan 2019 — Present*
@ *University of Colorado, Colorado Springs* *10 hrs/wk*
- **Educational Support:** Delivered lectures, held office hours, graded assignments/exams, and provided tutoring for approximately 160 students.
- **Adaptability:** Tailored teaching methods to accommodate diverse learning styles and classroom environments, effectively enhancing student understanding and engagement of complex cryptographic concepts. Regularly adapted and updated course materials to ensure relevance.
- **Self-Driven:** Proactively developed, refined, and updated course materials, implementing Python-based and JavaScript-based automation tools to significantly improve grading efficiency, accuracy, and overall workflow.

Graduate Research Assistant – Secure Networking Protocols and Analyses *May 2020 — Present*
@ *University of Colorado, Colorado Springs* *40 hrs/wk*
- **Problem-Solving:** Collaborated to enhance networking protocols by identifying and addressing vulnerabilities in Bitcoin, Ethereum, QUIC, and 5G Communication Systems, and developing robust countermeasures against novel threats.
- **Data Analysis:** Analyzed network behaviors and utilized machine learning in connectivity/behavior estimation, and anomaly detection.
- **Detail-Oriented:** Conducted rigorous analyses and published findings in reputable conferences and journals, including a responsible disclosure.

Teaching Assistant – *CS 4300 / 5300*, Advanced Software Engineering *Aug 2023 — Dec 2023*
@ *University of Colorado, Colorado Springs* *10 hrs/wk*
- **Leadership:** Led 48 students across 8 teams in software development and testing best practices using CI/CD, build/test automation, and test-driven/behavior-driven development. Used GitHub and ZenHub for project management simulating real-world development scenarios.
- **Communication:** Provided support during office hours and was responsible for grading assignments, projects, presentations, and write-ups.

NSF JROTC Mentor – Network Fundamentals *May 2021 — Aug 2022*
@ *United States Air Force* *30 hrs/wk*
- **Mentorship:** Instructed 36 high school cadets from diverse backgrounds on *Network Fundamentals* and individual research projects, preparing students for the CompTIA certification with complementary exam vouchers upon course completion.

Project Mentor – Crypto for Charity *Jun 2021 — Aug 2022*
@ *Silicon Valley Cyber Security Institute* *20 hrs/wk*
- **Innovation:** Guided a series of projects with undergraduate students, with a focus on blockchain and cryptocurrency use for charitable causes.

Teaching Assistant – *CS 4910*, Introduction to Computer Security *Aug 2018 — Dec 2021*
@ *University of Colorado, Colorado Springs* *10 hrs/wk*
- **Security Instruction:** Led interactive classes, graded exams and coursework, and provided tutoring for approximately 80 students.

Undergraduate Research Assistant – Blockchain Technology *Jun 2018 — May 2020*
@ *University of Colorado, Colorado Springs* *40 hrs/wk*
- **Technical Proficiency:** Developed simulation tools for analyzing blockchain network behaviors and enhancing understanding of the system.
- **Research Initiative:** Identified and addressed vulnerabilities in blockchain systems, notably within Bitcoin to advance its network security.
- **Project Leadership:** Managed research activities leading to conference publications, and secured a Ph.D. scholarship post-graduation.

AWARDS & HONORS

- Scholarship for Service (SFS), Ph.D. in Security — *Aug 2020 — Present*
- O'Neil Graduate Research Assistantship Student Grant — *Aug 2022 — Aug 2023*
- Best Paper Award, ICBC 2021 & CCNC 2023 — *May 2021 — Jan 2023*
- NSA Graduate Cybersecurity Scholarship — *Jan 2022 — Aug 2022*
- Top Scholar Award, UCCS Mountain Lion Research Week — *Received — Nov 2020*
- Security Scholar at RSA Conference (Scholarship Recipient) — *Received — Mar 2019*
- SkillsUSA State-Wide Competition for Web Design — *Received — Apr 2015*

MEMBERSHIPS & EXTRACURRICULARS

Organized and Hosted Six Capture the Flag (CTF) Competitions — *Sep 2021 — Mar 2022*

- Conceptualized and led six jeopardy-style CTF competitions focused on computer science and cybersecurity, involving 91 participants.
- Administered entry and exit surveys, gathering feedback and insights that contributed to two publications showcasing CTF's educational value and collaborative team building through skill matching and real-time score synchronization.

Volunteer Coordinator and Presenter at Silicon Valley Cybersecurity Institute (SVCSI) — *Sep 2022 — Present*

- Actively involved in managing the annual SVCSI event, a precursor to the SVCC conference. Responsible for handling the setup for approximately 100 participants, ensuring smooth event flow, efficiently addressing inquiries, and presenting my research at the conference.

Mountain Lion Cyber Club at UCCS — *Aug 2022 — Present*

- Member of the weekly cybersecurity club, covering topics in network security, ethical hacking, digital forensics, and other skill development opportunities. Engaged in diverse activities including interactive training sessions, capture-the-flag events, and cybersecurity workshops.

Artist: Passionate about painting, having created several distinct pieces that serve as a creative and reflective balance to my technical work.

OPEN-SOURCE CONTRIBUTIONS & NOTABLE PROJECTS

5G Protocol Security Enhancement — *Mar 2022 — Present*

- Directed team implementation to enhance 5G security through novel base station authentication methodologies.
- Leveraged srsRAN with 5G standardization and 4G/5G SDR equipment for prototype development and experimentation.

Blockchain Network Simulator — *Oct 2019 — Present*

simewu.com/blockchain-simulator

- Developed a versatile proof-of-work network simulator, ideal for modeling diverse network environments for performance/security logging.
- Since its introduction at ICDCS 2021, the simulator has *ranked among the top five Google searches for "blockchain simulator" globally*.

Bitcoin & Ethereum Network Logging Enhancements — *Sep 2018 — Present*

- Engineered real-time logging and analysis tools in Bitcoin Core and Ethereum Virtual Machine (EVM), notably improving P2P node tracking, data storage, and system interaction capabilities; discovered and disclosed a networking vulnerability to the Bitcoin Core developers.
- Collaborated with Ethereum developers on tools analyzing opcode executions, revealing insights on contract-specific metrics and performance.

AES & DES Calculators — *Mar 2018 — Aug 2020*

simewu.com/aes & simewu.com/des

- Converted cryptographic algorithms into dynamic and user-friendly formats, facilitating cryptography understanding and education.
- The DES Calculator has *ranked among the top five Google search results for "des calculator" globally.*

RESEARCH PUBLICATIONS

26 Published, 3 Accepted, 3 Under Peer Review

- "Securing Post-Quantum DNSSEC Against Fragmentation Mis-Association Threat" — *ICC 2024*
- "SEAL: Secure and Efficient Authentication using Linkage for Blockchain Network" — *ICIN 2024*
- "Positive Reputation Score for Bitcoin P2P Network" — *CCNC 2024**
- "Base Station Gateway to Secure User Channel Access at the First Hop Edge" — *ComNet 2024*
- "Post-Quantum QUIC Protocol in Cloud Networking" — *EuCNC 2023*
- "Version++: Cryptocurrency Blockchain Handshaking with Software Assurance" — *CCNC 2023**
- "Version++ Protocol Demo for Cryptocurrency Blockchain Handshaking with Software Assurance" (Best Demo Award) — *CCNC 2023**
- "Robust P2P Networking Connectivity Estimation Engine for Permissionless Bitcoin Cryptocurrency" — *ComNet 2022*
- "QUIC Protocol with Post-quantum Authentication" — *ISC 2022*
- "A Machine Learning Approach to Anomaly Detection based on Traffic Monitoring for Secure Blockchain Networking" — *TNSM 2022*
- "The Security Investigation of Ban Score and Misbehavior Tracking in Bitcoin Network" (Responsible Disclosure) — *ICDCS 2022**
- "Greedy Networking in Cryptocurrency Blockchain" — *IFIP SEC 2022*
- "Lightweight Code Assurance Proof for Wireless Software" — *WiSec 2022**
- "Post-Quantum Cipher Power Analysis in Lightweight Devices" — *WiSec 2022**
- "Designing and Using Capture the Flag for Coordination and Interaction in Engineering Education" — *EDUCON 2022*
- "Capture the Flag for Team Construction in Cybersecurity" — *ArXiv 2022*
- "Lightweight and Identifier-Oblivious Engine for Cryptocurrency Networking Anomaly Detection" — *TDSC 2022*
- "Towards Securing Availability in 5G: Analyzing the Injection Attack Impact on Core Network" — *SVCC 2021*
- "A Machine Learning Approach to Peer Connectivity Estimation for Reliable Blockchain Networking" — *LCN 2021*
- "Enhancing Cybersecurity Education and Workforce through Colorado-Washington Security Scholar Program" — *CCSC 2021*
- "Performance Characterization of Post-Quantum Digital Certificates" (Invited Paper) — *ICCCN 2021*
- "Proof-of-Work Network Simulator for Blockchain and Cryptocurrency Research" — *ICDCS 2021**
- "Robust P2P Connectivity Estimation for Permissionless Bitcoin Network" — *IWQoS 2021*
- "Security Comparisons and Performance Analyses of Post-Quantum Signature Algorithms" — *ACNS 2021**
- "Security Analyses of Misbehavior Tracking in Bitcoin Network" — *ICBC 2021**
- "Anomaly Detection based on Traffic Monitoring for Secure Blockchain Networking" (Best Paper Award, Finalist) — *ICBC 2021**
- "Dynamic Power Control for Rational Cryptocurrency Mining" — *CryBlock 2020*
- "Anti-Withholding Reward System to Secure Bitcoin Mining Pools" — *CVCBT 2019*
- "Uncle-Block Attack: Blockchain Mining Threat Beyond Block Withholding for Rational and Uncooperative Miners" — *ACNS 2019**

** Indicates conferences recognized for their selectivity (acceptance rate below 25%).*